



# Lucas Furer

## Computer Science graduate

A detail-oriented and enthusiastic MSc Computer Science graduate with a passion for physics-based simulations, computer graphics and open-source principles. I am an eager learner who enjoys collaborating with others and exploring new technologies. Experienced with C++, Java, GLSL and Python.

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 Delft, NL

### Education

- M.Sc. Computer Science, TU Delft** 2023-2026
- **Thesis:** **FMM-t-SNE** Implemented a fast multipole N-body solver and integrated it into t-SNE. Developed in C++ with OpenGL visualization
  - **Grade:** 8.5
  - **Notable courses:** Image Processing, Graphics and Animation, Geometric data Processing
- B.Sc. Computer Science, TU Delft** 2019-2023
- **Thesis:** **Soft Peaks** Studied how the distribution of ray reflections evolve as smoothing is applied to microstructures. Implemented in C++ with results rendered to .ppm
  - **Grade:** 7.5
  - **Minor:** Security, Safety & Justice
  - **Notable courses:** Linear Algebra, Big Data Processing, Computer Graphics

### Experience

#### Internship

- Feedback Analytics, Software Engineer** 2022
- Developed a visualization tool for user-created questionnaires using Angular 13
  - Lead a five-person team, collaborating closely with the client to gather requirements

#### Projects

- Blender Addon, Author** 2026
- Created a Blender addon that generates terrain using simulated hydraulic erosion
  - **Source Code:** [https://github.com/LucasFurer/my\\_blender\\_addon](https://github.com/LucasFurer/my_blender_addon)
- Independent Game Development, Software Engineer**
- Organized a four-person team to create a game using Unity and C#
  - Developed procedural terrain generation using Perlin noise and the Marching Cubes algorithm

### Skills

#### Programming

**C++**  
Libraries: Boost, Eigen, Fastor, GLM, OpenGL, GLFW

**C, Python, C#, Java, GLSL, CUDA, Unity**  
Libraries: Numpy, Matplotlib, Pandas, OpenTSNE, Sklearn

**x86-Assembly, HTML, CSS, Javascript, Scala, Rocq, Haskell, SQL, MiniZinc**

#### Tools

**CMake, Docker, Linux, Bash, Git, Blender, LaTeX**

#### Language

**Dutch, Native      English, Fluent**

### Extracurricular Activities

- D.S.M.G. Krashna Musika, Bass singer** 2024-2026
- Performed with a classical music student association in concerts across Europe
  - Developed teamwork, discipline and performance skills through rehearsals and live performances
- D.S.B.V. Yoroshi** 2024-2025
- Practised judo at a student budosport association

### Interests

Playing piano, creating games/simulations, running, bouldering